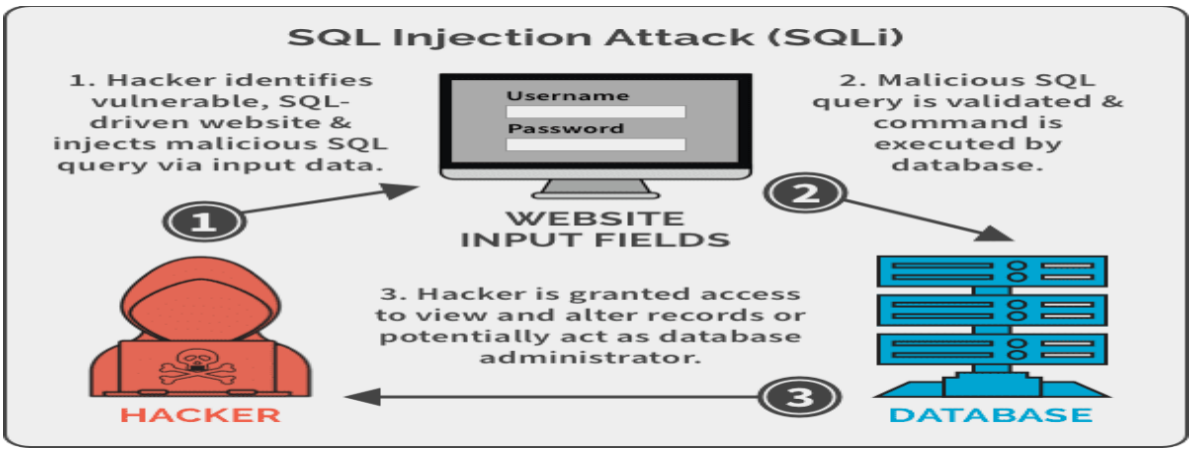
SQL INJECTION ATTACK:

What Is SQL Injection:

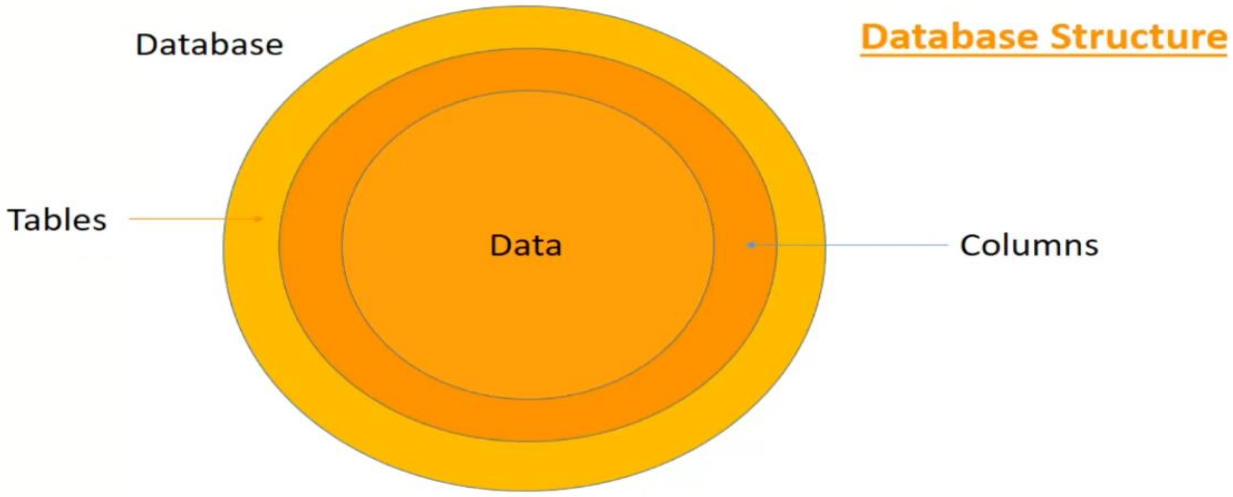
SQL Injection (SQLi) is a cyberattack technique where a hacker inserts malicious SQL code into an input field (like a login form or search box) to manipulate or access a database illegally.



In this section we are going to take a look at ....

- Database Attack Techniques for Database.
- Database Attack Tools in Kali Linux O.S

Structure of the database:



https://screenrec.com/share/h6GVUDBxoe

Database Attack Information:

- Gather Sensitive Information from Database
- Information like Username, Password, E-mail, Address can be gathered.
- Database Assessment tools are available in kali Linux O.S. (like SQL map, JSQL).

Website Database Hacking Using SQL map Tool.

SQL map – Step-by-Step:

What is SQL Map:

- SQL Map is a command-line tool
- It is used to automate SQL injection testing
- It helps in identifying databases, tables, columns, and data

1. Initial Preparation

- Open Google
- Open any website
- Do not perform on live websites
- Only test on:
    - Your own website
    - Test or practice websites

2. Finding the Login Page

- Open the website
- Go to the sign-in / login page
- Identify where:
    - Username field exists
    - Password field exists
- Copy the URL of the login or signup page

3.Identifying Source Links

- Open a new Google page
- Use search technique with:
    - **Site:website URL/ php?id=**

    This shows multiple internal links from the website

4.Open the terminal:

- **Aim is to find – Database – table – columns and data.**

Step1: to find the database:

Sqlmap -u websitelink – - dbs

Step 2: After getting the databases in the website find the tables in the database:

Sqlmap -u websitelink – D database name - –tables

Step 3: After getting the tables in the particular database then find the columns:

Sqlmap -u websitelink – D database name –T table name - - columns

Step 4: After getting the columns find the sensitive information from the columns given below like uname, password, email.

Sqlmap -u websitelink – D database name – T table name -C columns name –dump

5.Using Retrieved Credentials:

- Go back to the admin or login page
- Enter:
    - Retrieved username
    - Retrieved password
- Login is successful
- User dashboard or credentials become visible

https://screenrec.com/share/oxL4adsrOX

Website Database Hacking Without Using any Tool.

Hacking a data base without any tool first 3 steps are same as hacking using tool only:

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2. Finding the Login Page

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- Go to the sign-in / login page
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  - Username field exists
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- Use search technique with:
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This shows multiple internal links from the website

AFTER THAT FROM HERE THE MAIN PART STARTS:

To Check that the website is vulnerable is sql injection put this url and check and if any waring comes then it is vulnerable to sql.

Warning: mysql_fetch_array() expects parameter 1 to be r
boolean given in /hj/var/www/artists.php on line 62

HOW TO CHECK: website url.php?uid=1'

Example: http://testphp.vulnweb.com/artists.php?artist=1'

4. After checking the website is vulnerable then check trail and error method for finding columns.

To check: website url.php?uid=1 order br 5 --

Important note:

In order by 5 – if the page has not change then use different integer like 3,4,6,7, etc.

Still the pages changes and gives us with no of columns.

https://screenrec.com/share/rlZRVFyLWE

Create Dictionary Using Crunch Commands:

What is Crunch?

Crunch is a wordlist generation tool that comes pre-installed in Kali Linux.
It's commonly used in penetration testing to create custom password lists for tools like air crack-ng, hydra, john, etc.

Usage or syntax: crunch <min> <max>.

Example:

root@kali: ~# crunch 3 8 liger -o /root/Desktop/liger.txt

root@kali: ~# crunch 5 8 liger@$ -o /root/Desktop/liger.txt

where **-o:** output

/ root/Desktop/liger.txt: where the out is saved